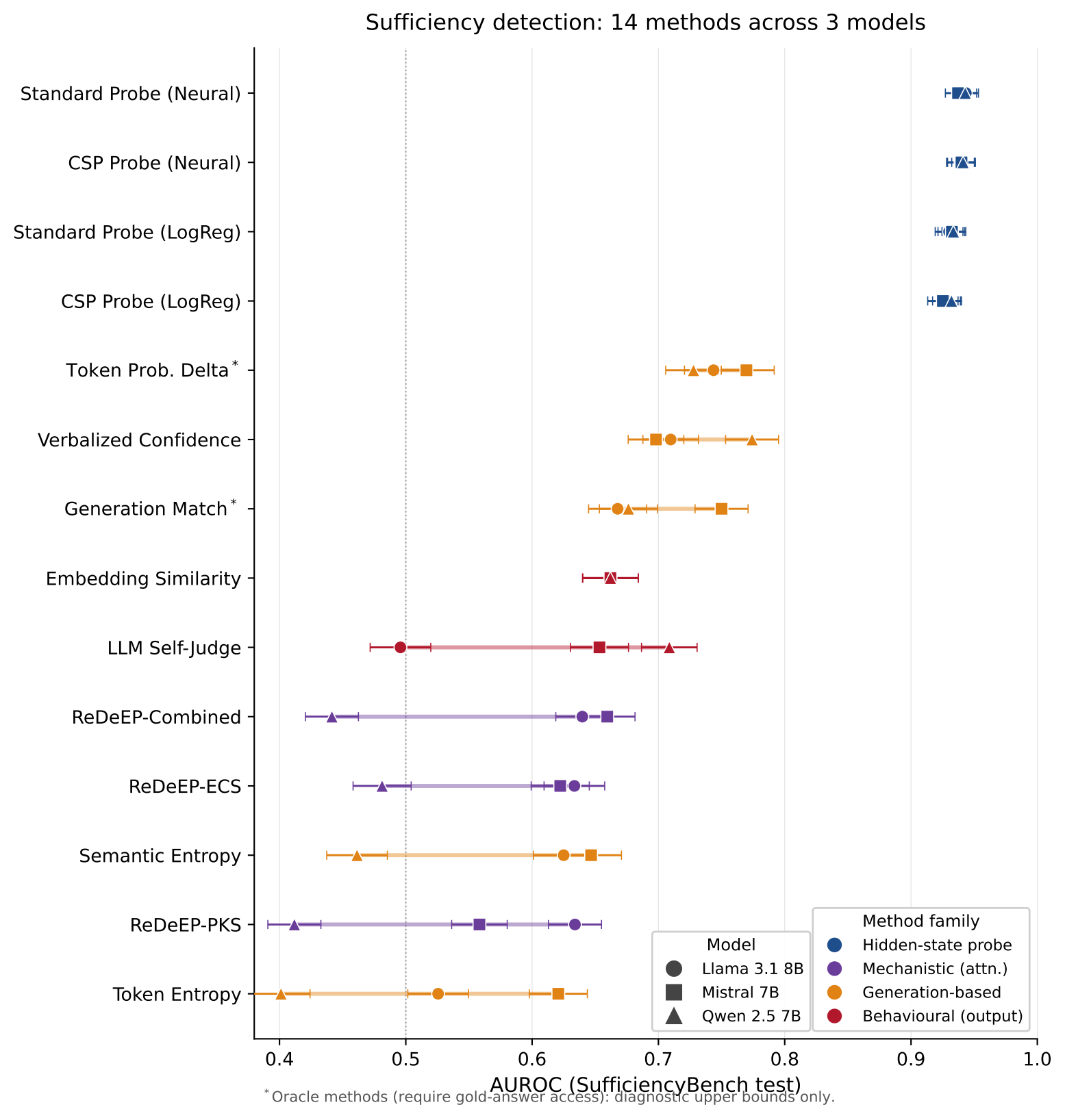
Sufficiency detection: 14 methods across 3 models
Standard Probe (Neural)
CSP Probe (Neural)
Standard Probe (LogReg)
CSP Probe (LogReg)
Token Prob. Delta*
Verbalized Confidence
Generation Match*
Embedding Similarity
LLM Self-Judge
ReDeEP-Combined
ReDeEP-ECS
Semantic Entropy
ReDeEP-PKS
Token Entropy
0.4
0.5
0.6
0.7
0.8
0.9
1.0
AUROC (SufficiencyBench test)
Model
Llama 3.1 8B
Mistral 7B
Qwen 2.5 7B
Method family
Hidden-state probe
Mechanistic (attn.)
Generation-based
Behavioural (output)
*Oracle methods (require gold-answer access): diagnostic upper bounds only.